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A Novel Plus Disease Detection Model for a Deployable Retinopathy of Prematurity Screening System with Deep Convolutional Neural Network Ensembles

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Abstract

Retinopathy of prematurity (ROP) is a serious vascular condition affecting the retina of premature infants. It is characterized by the abnormal growth of blood vessels, which, if left untreated, can lead to permanent blindness. With the increased rate of preterm babies and the unavailability of sufficient ROP experts, screening of ROP to decide whether the baby requires treatment or not has become challenging. The presence of Plus disease, characterized by unusual tortuosity and dilation of blood vessels in the posterior retina, is a critical indicator for identifying ROP cases that require treatment. The primary objective of this work is to develop a generalizable Plus disease classifier with 100% sensitivity, enabling clinicians to conduct large-scale screening. A deep convolutional neural network (CNN)-based ensemble classifier is designed for the automatic detection of Plus disease. A state-of-the-art CNN model of EfficientNetV2-S has been trained independently on two distinct Plus/No Plus datasets to generate independent predictions, which are then combined through ensemble learning to enhance overall accuracy. The proposed model achieves 100% sensitivity (zero false negatives), with a very high specificity value, which is the requirement for a deployable ROP screening system. Also, for optic disc (OD) centered infant fundus images, the system could produce zero-misclassification, which resulted in a new benchmark for Plus disease classification. The model prevents missed ROP diagnoses and reduces expert review of healthy cases. The developed model is highly reliable and well-suited for deployment in clinical screening environments, where accurate and early detection is critical.

Keywords Retinopathy of prematurity screening · Plus disease · Convolutional neural networks · Ensemble models · Classification

1 Introduction

Retinopathy of prematurity (ROP) is a vitreo-retinal disorder associated with abnormal retinal vascular development in premature infants. In advanced stages, the disease may lead to retinal detachment due to the formation and contraction of fibrovascular membranes within the vitreous cavity and along the retina.

Advances in neonatal medical care have significantly improved survival rates among preterm infants, particularly in developing nations such as India, but have also led to a rise in ROP cases globally. Each year, approximately 14.38 million preterm infants are born worldwide, with India accounting for 23% of these cases. Recent studies in 2023–2025 indicate that the incidence of ROP among screened preterm infants in India ranges from 16 to 42%, depending on the region, gestational age, and birth weight. The proportion of infants developing treatment-requiring

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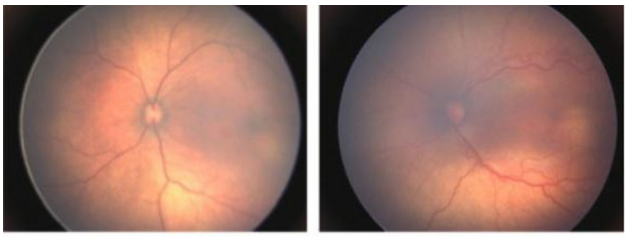


Fig. 1 Retinal images of premature infants taken from the KIDROP dataset: **a** normal **b** with plus disease

(severe) ROP has been reported between 6 and 9% in tertiary care studies [1]. Premature infants born before 32 weeks of gestation or weighing under 2000 g face a higher risk of developing ROP. In India alone, around 490,000 infants are born annually at a gestational age of less than 32 weeks, and at least 5000 of them require treatment for ROP. Additionally, factors such as oxygen therapy and its duration play a critical role in the onset and severity of ROP.

ROP severity is classified into five stages as per the International Classification of Retinopathy of Prematurity (ICROP) [2]. Stage 1 shows a thin line demarcating vascular and avascular retina, while Stage 2 features a broader ridge. Stage 3, marked by neovascularisation and a velvety ridge, is the threshold for medical intervention. Stage 4 involves partial retinal detachment, progressing to total detachment in Stage 5. Plus disease, indicative of severe ROP requiring treatment, may appear from Stage 3 onward. Figure 1 presents digital fundus images comparing a normal infant retina with that of an infant affected by Plus disease. The differences in blood vessel characteristics between the two are distinctly noticeable, serving as a key indicator of disease progression to the threshold stage. Retinal images with Plus disease exhibit blood vessels that are noticeably more tortuous and dilated compared to those in a healthy retina. If the disease is not diagnosed and treated at the right stage, it leads to retinal detachment and permanent blindness.

Screening a large number of premature babies for ROP to decide the severity of the disease is the major challenge in ROP analysis. Out of them, screening to identify the non-diseased cases and treatment-requiring cases remains the most important one. This two-way classification, identified by the presence of Plus disease, is challenging due to the basic nature of the identification benchmarks, such as vessel tortuosity and dilation. The decisions by ROP experts often lead to inter-clinician variations [3]. The scarcity of trained ROP experts makes the problem twofold and demands an automated screening system with 100% sensitivity, to ensure that no diseased babies are left untreated. This is a critical requirement, since these babies are under neonatal care in the first few weeks after birth [4].

The classification of Plus disease in ROP has been a focal point for researchers employing various machine learning and deep learning techniques. Traditional machine learning approaches [5] typically involve segmenting blood vessels, extracting handcrafted features, and using these features for classification [6]. However, the accuracy of such methods is highly dependent on precise segmentation and feature extraction. Errors during segmentation [7] can propagate through the classification pipeline, compromising the reliability of results. To address these limitations and mitigate the bias associated with handcrafted features, recent efforts have shifted toward deep neural networks. Convolutional neural network (CNN) based approaches, which enable automated feature extraction and improved diagnostic accuracy, are the best approach for efficiently classifying Plus and No Plus cases.

For instance, Worrall et al. [8] employed a modified GoogLeNet classifier on a dataset of 1459 images to categorize ROP into diseased or healthy classes, achieving 95.4% sensitivity but compromising on accuracy (93%) and specificity (94.6%). Wang et al. [9] analyzed 2361 ROP images using the id-Net architecture, reporting a sensitivity of 96.64% but an accuracy of only 95%. Mao et al. [10] classified 3311 images into Plus, Pre-Plus and No Plus categories using DenseNet, achieving 93.31% accuracy and 95.1% sensitivity. Tong et al. [11] utilized a dataset of 36,231 fundus images annotated by 13 retinal experts. They implemented two convolutional neural networks: a 101-layer ResNet and Faster R-CNN, achieving 90% accuracy and 93.2% specificity, but a lower sensitivity of 77.8% for ROP severity classification. Lei et al. [12] introduced an attention module with ResNet-50 for the classification of 17801 images (8090 diseased and 9711 healthy), achieving 98.03% sensitivity but a specificity of only 94.55%. Zhang et al. [13] developed a two-network system for diagnosing Aggressive Posterior ROP (AP-ROP). Network 1 detected the presence of ROP, while Network 2 distinguished regular ROP from AP-ROP. Their system, trained on 6867 images and tested on 3654, achieved an accuracy of 93.09%, but 87.70% sensitivity.

Nazar et al. [14] proposed a fusion-based deep learning framework for automated ROP diagnosis using 2776 fundus images. Their approach combined ResNet50, DenseNet161, and Efficient-NetB5 architectures for ROP/Normal classification, achieving an accuracy of 90.28%. Elizabeth et al. [15] developed a deep learning-based ROP screening system employing a customized ResNet-50 architecture integrated with a deep neural network classifier. Their framework performed both ROP/Normal classification and Stage 2/Stage 3 severity classification, attaining 92.8% accuracy and 94.9% sensitivity. Raja Sankari et al. [16] introduced a novel Quantum Vision Transformer (QMViT) model for Plus disease

classification in ROP using 2,400 retinal fundus images comprising 2000 ROP and 400 Normal cases. The proposed model achieved a classification accuracy of 96.67%. Agrawal et al. [17] utilized retinal images from premature infants, including 61 Pre-Plus, 65 Plus, and 65 Normal samples. Deep features extracted using pretrained InceptionV3, Xception and ResNet152 models were classified using a DNN classifier, resulting in a testing accuracy of 87% for Pre-Plus, Plus, and Normal classification. Nasser et al. [18] analyzed nearly 1700 RetCam images obtained from 141 preterm infants and applied preprocessing techniques such as CLAHE and Retinex enhancement. Several CNN architectures, including MobileNet, ResNet-18, ResNet-50, and DenseNet-121, were investigated for treatment-requiring ROP detection. Among them, MobileNet combined with CLAHE preprocessing achieved the best performance with 91.39% accuracy and 94.9% sensitivity. Esraa et al. [19] proposed the ROPDeepX framework based on ResNet50 and EfficientNet-B4 for the classification of Normal, Pre-Plus, and Plus disease categories. The model was evaluated on 7637 images, achieving an overall classification accuracy of 96.9%.

Though many works are in line with the ROP and Plus Disease detection, one of the major clinical requirements of correctly identifying all the treatment-requiring cases (Presence of Plus disease) and the discharge cases (No Plus) is not properly addressed in the literature. Many architectures use a large amount of training data; however, the false classification is still significant, which is reflected in the sensitivity and specificity values. The data imbalance, where the number of diseased cases is low, can also be a reason for achieving lower values for specific evaluation matrices. One of the major challenges while using deep networks such as CNN, in medical diagnosis is the data scarcity. In many of the classification problems, a lack of a sufficient number of diseased cases leads to an unbalanced training set, which causes biased classification decisions. In this work, we try to address this challenge in two ways. One is by employing a transfer learning approach, where pretrained CNN architectures are used as base models to train the ROP data. Another one is by employing a dual-stage data augmentation technique, which relies on efficient enhancement methods that generate data to enable the network to learn features more effectively.

The primary aim of the work was to use the transfer learning approach and identify an existing pretrained CNN architecture that, when trained on the ROP dataset, will classify the diseased cases with 100% sensitivity. However, identifying a single such network, even with augmented data, was found to be difficult. Hence, performance analysis of the existing pretrained CNN architectures on the ROP data was done to identify the best-performing architectures. Various ensembling approaches were experimented with using the selected network models. Analysis of the available data with various deep networks revealed the possibility of ensembling using a bagging approach for the effective learning of the existing deep net-works. Hence, an ensemble CNN architecture is proposed by effectively combining the predictions from models trained on different datasets, which achieved the target of 100% sensitivity. The key contributions of this study are as follows:

- *Dual-stage augmentation and enhancement pipeline:* The proposed framework incorporates a dedicated preprocessing and dual-stage data augmentation strategy specifically designed for infant retinal fundus images to address limited data availability and image variability associated with ROP screening.
- *An efficient bagging ensemble framework with optimal data segregation:* EfficientNetV2-S models were independently trained on two distinct Plus/No Plus datasets to learn dataset-specific representations, and their outputs were combined using a Soft Voting strategy.
- *Clinically driven optimization objective:* Unlike many existing studies that primarily optimize overall accuracy, the proposed framework specifically targets 100% sensitivity (zero false negatives), which is critically important in ROP screening, as missed diagnoses may lead to irreversible blindness.
- *Benchmark-level OD-centered classification performance:* The proposed framework achieved zero misclassification for Optic Disc-centered retinal fundus images, demonstrating strong screening capability and benchmark-level classification performance.
- *Deployment-oriented screening framework:* The developed system is designed for practical clinical screening applications by reducing expert workload while maintaining highly reliable dis-ease detection performance.
- To the best of our knowledge, this is the first study to propose a deep learning-based ensemble framework specifically for Plus disease classification in ROP screening.

The remainder of this paper is organized as follows. Section 2 describes the proposed methodology for Plus disease classification. Section 3 presents the experimental results. Section 4 discusses the obtained findings and provides a comparative analysis with previously reported state-of-the-art methods. Finally, conclusions are drawn in Sect. 5.

2 Methods

This section describes the data augmentation strategy proposed as a solution to mitigate the data deficiency issues in

its first part. The second part gives the details of the analysis done to arrive at the proposed ensemble classifier for the Plus/No Plus disease classification.

A review of the literature reveals that most studies employing deep neural network architectures for ROP diagnosis typically rely on large datasets. However, a major challenge in medical image analysis in general is the limited availability of diseased cases. This leads to a predominance of normal cases, which results in data imbalance: an issue that significantly impacts the network training. Another critical limitation is the absence of publicly available datasets; most existing studies rely on private datasets, making a fair comparison of models difficult. This work tries to address the data limitations through various augmentation techniques, which have not been well studied in the literature so far.

The primary focus of this study is on the binary classification of Plus versus No Plus ROP images, exploring the transfer learning strategy using the pretrained CNN architectures with a limited dataset. The analysis done on the existing pretrained models guided us to an ensemble model, which helped to meet the target of 100% sensitivity by combining predictions from multiple models [20, 21].

A simplified block diagram illustrating the proposed Plus disease detection is presented in Fig. 2. As illustrated in the block diagram, the proposed Plus disease classification system consists of three main stages: (i) Dual-stage data augmentation, (ii) Selection of the best pretrained CNN models, and (iii) Ensembling-based classification. The dual-stage data augmentation technique employed in the first stage to expand the training set includes image enhancement methods and geometric transformations. Including enhancement methods to augment the data aims to increase data variability and improve the generalization ability of the models. In the second stage, the best pretrained models are selected based on the accuracy in classifying the diseased cases. Here, the entire dataset, segregated into train, validation, and test, is used with augmentation applied to the train and validation sets. The ensembling strategy adopted to enhance classification robustness and to meet the screening system requirements is described in the third stage. The data were segregated into optic disc (OD) centered images and Non-Optic Disc centered ones. The best performing pretrained model obtained by training on the entire dataset is trained on the augmented dual dataset, and their predictions are integrated using a soft voting mechanism. This ensemble approach leverages the complementary strengths of individual models to improve overall performance. The details of various stages in the complete workflow are elaborated in the following sections.

2.1 Dual Stage Data Augmentation

Data augmentation is widely used for increasing the number of images for training deep neural networks, especially when there is a scarcity of data for training the network [22]. The proposed data augmentation framework consists of a dual-stage data augmentation strategy to increase image heterogeneity by preserving prognostic characteristics of the image. All augmentation procedures were performed only after patient-wise dataset separation to prevent any possibility of data leakage between the training, validation, and testing subsets. Furthermore, all augmented samples generated from a particular patient were restricted to the same subset as the corresponding original image to ensure complete patient-level isolation during model development and evaluation.

The proposed work uses two major approaches for data augmentation. The first approach uses various enhancement methods on the original images for data augmentation. The second approach uses the most common data augmentation strategies, such as geometric and coordinate transformations on the original as well as the data augmented using the first approach.

Data augmentation using Image enhancement methods: The primary objective of the first augmentation stage is to improve the visibility of diagnostically relevant retinal structures, particularly retinal vasculature and vessel tortuosity patterns associated with Plus disease. In general, enhancement methods that improve image appearance while preserving clinical information can be incorporated into the augmentation framework. The enhanced retinal images assist the network in learning discriminative features more effectively and improve convergence during training.

In the proposed framework, image enhancement methods previously proposed by the authors [23], including filtering operations, brightness enhancement, contrast adjustment, and detail boosting techniques based on Gaussian and Guided filters, were utilized for augmentation. Figure 3 illustrates the block schematic of the enhancement methods that are used to extract features as employed in [23]. The result of various stages of the flow diagram 3 (a) and 3 (b) for a sample image is given in Figs. 4 and 5, respectively. Initially, the green channel plane was extracted from the original retinal image because retinal vascular structures are more prominently visible in the green channel. Guided filtering and Gaussian filtering operations were then separately applied to enhance vessel visibility and suppress noise. Subsequently, brightness and contrast enhancement operations were performed, followed by detail boosting to improve the visualization of fine vascular structures. Finally, the enhanced green channel was recombined to generate the enhanced color retinal image.

The enhancement-based augmentation methods generated multiple clinically improved representations of the same retinal image while preserving the original diagnostic characteristics. The enhanced images exhibited clearer retinal vascular structures compared to the original images, thereby

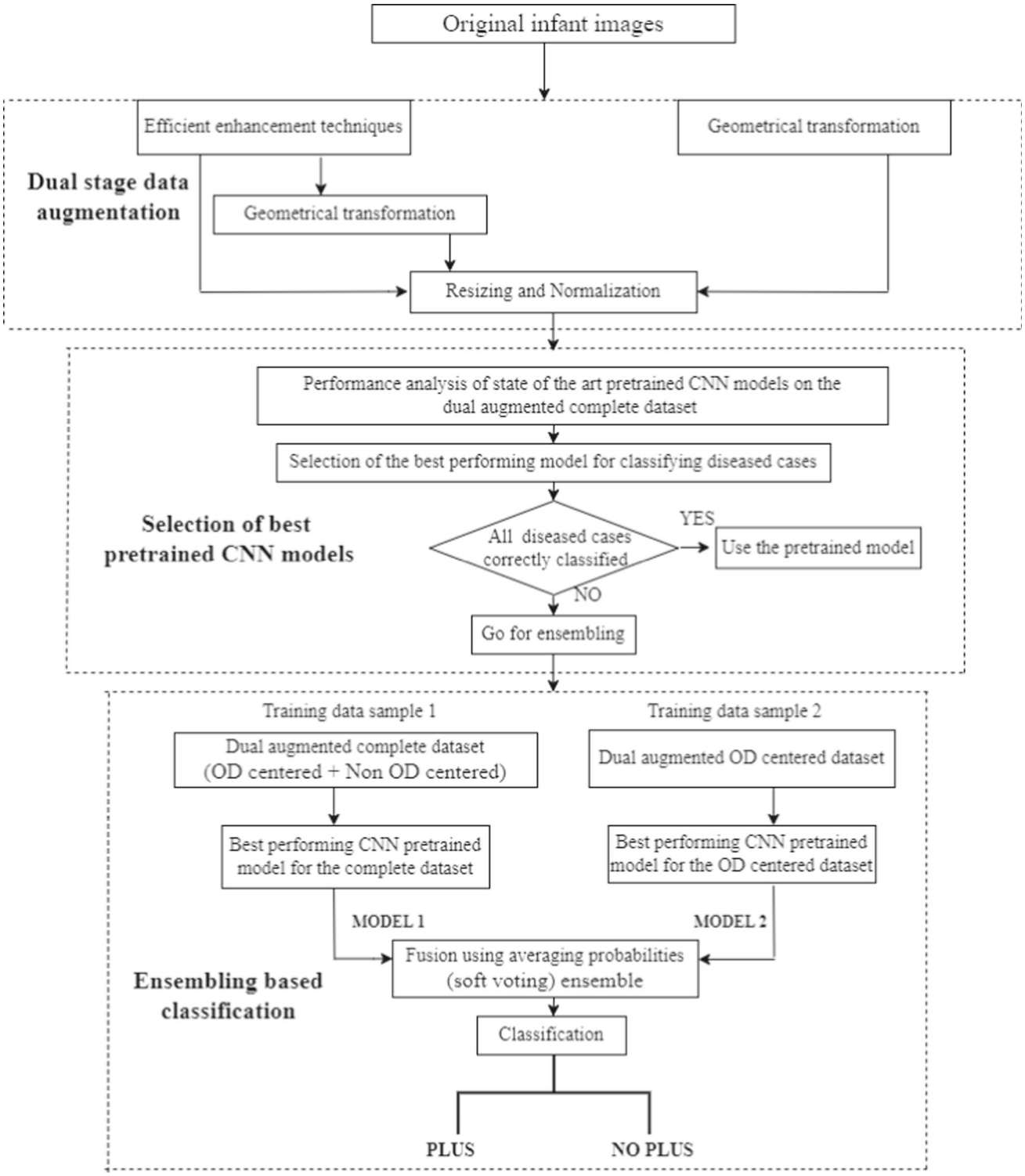


Fig. 2 Overview of the proposed Plus disease classification system

assisting the pretrained CNN models in learning disease-specific features more effectively.

Geometrical/Co-ordinate transformations: To further increase the number of training samples and improve model robustness, geometric and coordinate transformation-based augmentation techniques were employed in the second stage of the augmentation framework. Since retinal fundus diagnosis primarily depends on the identification of anatomical

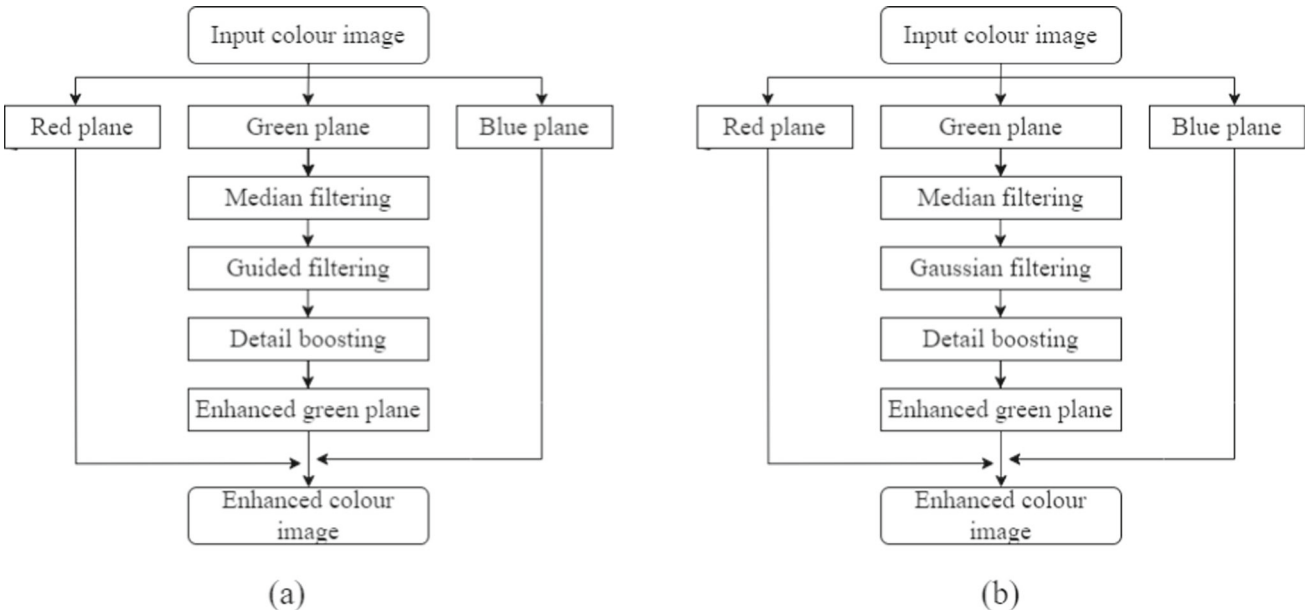


Fig. 3 Block schematic of image enhancement methods

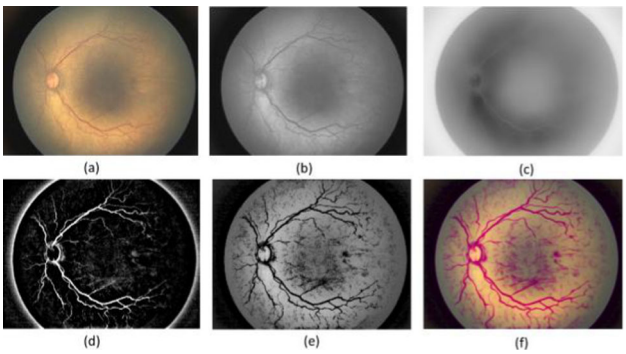


Fig. 4 Output stages of flow diagram 3 (a): **a** Original image **b** Green plane **c** Guided filter **d** Boosted details **e** Enhanced green plane **f** Enhanced color image

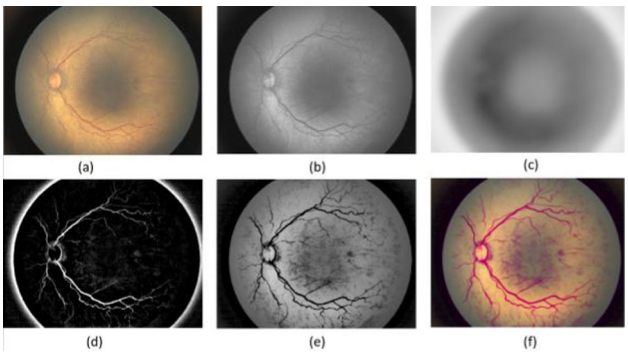


Fig. 5 Output stages of flow diagram 3 (b): **a** Original image **b** Green plane **c** Gaussian filter **d** Boosted details **e** Enhanced green plane **f** Enhanced color image

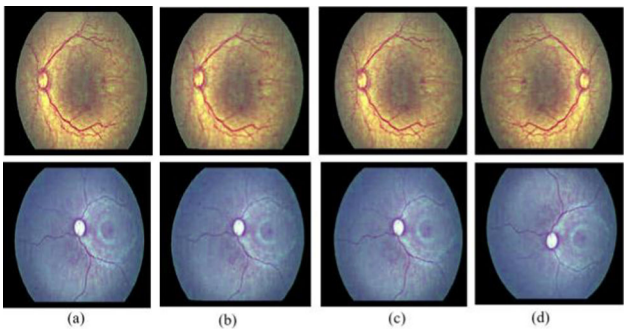


Fig. 6 Results of geometric and coordinate transformations on the enhanced images using method 3 (a): top row – images with Plus disease, bottom row – normal images. **a** Enhanced image **b** Rotated version **c** Translated version **d** Reflected version

structures such as retinal vasculature, vessel width, and tortuosity, label-preserving geometric transformations were selected.

The geometric augmentation operations included image translation, rotation, and reflection transformations. Translation operations were performed using pixel coordinate shifts of (–2, 3) and (–8, 8). Rotation transformations were applied at different angular orientations to improve rotational invariance of the network. Reflection transformations were also employed, where the input image was flipped with a probability of 50% in each dimension. These transformations were applied to both the original retinal images and the enhancement-based augmented images generated during the first augmentation stage. Figures 6 and 7 present the results of geometric and coordinate transformations applied to the enhanced images generated through the first stage of the

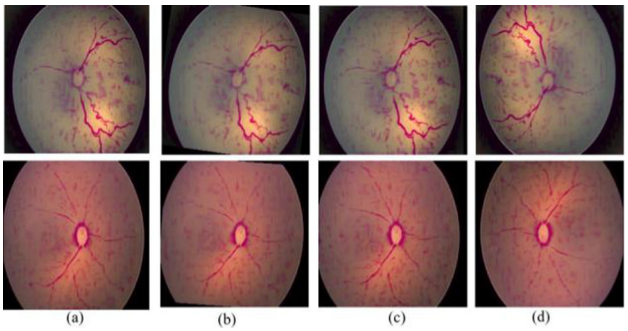


Fig. 7 Results of geometric and coordinate transformations on the enhanced images using method 3 (b): top row – images with Plus disease, bottom row – normal images. **a** Enhanced image **b** Rotated version **c** Translated version **d** Reflected version

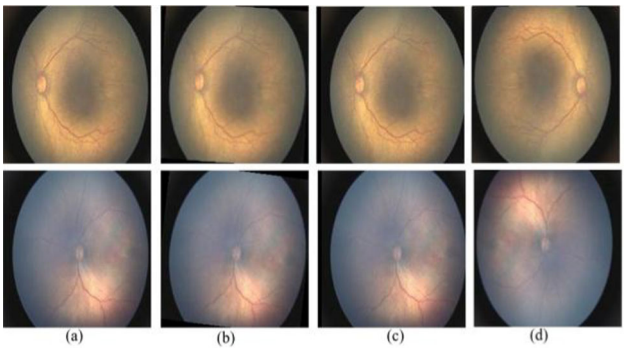


Fig. 8 Results of geometric and coordinate transformations on the original image: top row – images with Plus disease, bottom row – normal images. **a** Original image **b** Rotated version **c** Translated version **d** Reflected version

dual-stage data augmentation. Figure 8 illustrates the same transformations applied to the original images.

Multiple augmented samples were generated from each retinal image through sequential application of enhancement and geometric transformations. Following augmentation, all retinal fundus images were resized from the original resolution of 640 × 480 pixels to 224 × 224 pixels to satisfy the input size requirements of the pretrained CNN architectures. In addition, normalization was performed to maintain the input features on a similar scale and distribution, thereby improving model convergence and training stability [24].

The proposed dual-stage augmentation strategy provided a cost-effective approach for increasing dataset diversity without collecting additional clinical data. By improving image heterogeneity and preserving diagnostically relevant retinal structures, the augmentation framework enhanced model generalization capability, reduced overfitting, and improved the robustness of the proposed Plus disease classification system.

2.2 Selection of the Best Pretrained CNN Models

The dataset used in this study comprised 376 retinal fundus images acquired from 164 unique patients, including multiple images per patient. The images were collected from Narayana Nethralaya Eye Hospital, Bangalore, India, using RetCam3, a high-end pediatric wide-field digital imaging camera. Informed consent was obtained from the parents of all study participants.

To prevent data leakage and ensure an unbiased evaluation, a strict patient-wise separation strategy was employed, ensuring that images from the same patient were assigned exclusively to a single subset. Out of the total dataset, 222 images from 112 patients were allocated for model development and subsequently divided into training and validation sets using an 80:20 ratio at the patient level. This resulted in 178 training images from 89 patients and 44 validation images from 23 patients. The remaining 154 retinal fundus images from 52 unique patients were used as an independent test dataset, designated as Test Set 1 (T1), comprising 78 Plus and 76 No Plus images.

To address the limited dataset size and improve model generalization, the dual-stage data augmentation strategy described in Sect. 2.1 was applied to the training and validation datasets. Following augmentation, the number of training images increased from 178 to 3050 images, while the validation dataset increased from 44 to 762 images. The augmented validation dataset remained fixed throughout all experiments and was used exclusively for performance monitoring and model selection without contributing to network weight updates. The dual-stage augmented training and validation datasets are collectively referred to as Data Set 1 (D1) throughout this paper. The independent test dataset was intentionally kept completely unaugmented to ensure fair and unbiased evaluation of the proposed framework.

The first attempt in the classifier design was to use the transfer learning approach on the popular pretrained CNN models available in the literature [25]. So in the first phase, the performance of the various pretrained models was analyzed. In phase 2, a detailed analysis using the best performing networks in phase 1 is done to identify the most efficient ensemble framework that achieves the goal of 100% sensitivity and a very high specificity.

2.2.1 Performance Analysis of the Existing Pretrained CNN Models

For the identification of a suitable deep neural network model for Plus disease classification, this study employs transfer learning with existing pretrained CNN models such as EfficientNetV2-S [26], MobileNetV2 [27], DenseNet-201 [28], ResNet-50 [29], GoogLeNet [30], VGG-16 [31], VGG-19 [32], and AlexNet [33]. These models are fine-tuned

Table 1 Hyper-parameter settings of the experiment

Hyperparameter	Values
Learning rate	0.00001
Optimizer	Stochastic gradient descent momentum(sgdm)
Mini batch size	10
Maximum epochs	20
Loss function	Cross-entropy
L2 regularization factor	0.001
Learn rate drop factor	0.1000
Validation frequency	3
Training / Validation split	80:20

Table 2 Experimental setup

Parameters	Specifications
Language	Matlab R2023
CPU	Intel Core i7 3770
Memory	8 GB Dual-Channel DDR3
GPU	4 GB Nvidia DGX GPU A100
Disk	240 GB SSD Kingston

specifically for ROP image classification by initializing the base models with ImageNet weights as a backbone. The fine-tuning was done using a cross-entropy cost function and stochastic gradient descent (SGD) with a small learning rate of 1e-5 and a momentum value set to 0.9. To prevent overfitting, L2 regularization is applied. Since the dataset contains two class labels, the original output layer with 1000 units is replaced by a 2-unit output layer with a soft-max activation function. The training parameters used for the network are detailed in Table 1. The proposed models are trained on the entire training dataset and then evaluated on the validation dataset to determine the most efficient model.

Experimental Setup: All experiments were carried out using a machine with the configurations as in Table 2.

Evaluation Metrics: The performance metrics used for evaluation are Accuracy (Acc), Sensitivity/Recall (Sens), Specificity (Spec), Precision, F-measure, and AUC (Area Under the Curve) [34].

Experimentation: To further emphasize the contribution of the proposed dual-stage data augmentation strategy discussed in Section 2.1, an ablation study was performed using three different training settings: (a) without augmentation, (b) with only geometric augmentation, and (c) with the proposed dual-stage augmentation strategy incorporating both geometric and enhancement-based augmentation techniques. As previously described, patient-wise dataset separation was strictly maintained before augmentation to avoid data leakage. In all experiments, 80% of the data were used for training and 20% for validation. The number of images in each dataset configuration is summarized in Table 3.

The experimentally obtained validation accuracies of the advanced CNN models EfficientNetV2-S, MobileNetV2, DenseNet-201, ResNet-50, GoogLeNet, VGG-16, VGG-19, and AlexNet under different augmentation settings are presented in Table 4. All networks were trained using the three dataset configurations to evaluate the influence of augmentation on classification performance. The results clearly demonstrate that the proposed dual-stage augmentation strategy consistently improves the validation performance of all pretrained CNN models compared to both the original dataset and the dataset augmented using only geometric transformations. The improvement is mainly attributed to enhanced feature diversity, improved image representation, and reduced overfitting achieved through the proposed dual-stage augmentation framework.

Furthermore, the training and validation loss curves of the sample network, EfficientNetV2-S, trained using the three dataset configurations are illustrated in Fig. 9. As observed from the convergence behavior, the model trained using the dual-stage augmented dataset exhibits improved convergence and better generalization performance with reduced overfitting compared to the other two configurations, as shown in Fig. 9c. Therefore, the model trained using the complete dual-stage augmented dataset was selected for further evaluation and analysis.

The expanded validation results of the advanced CNN models EfficientNetV2-S, MobileNetV2, DenseNet-201, ResNet-50, GoogLeNet, VGG-16, VGG-19, and AlexNet trained using the complete dual-stage augmented dataset D1 are listed in Table 5. To improve the clarity of presentation and to have a more compact representation, the pretrained models EfficientNetV2-S, MobileNetV2, DenseNet-201, ResNet-50, GoogLeNet, VGG-16, VGG-19 and AlexNet trained on the complete dataset D1 are coined as M_1, M_2, M_3, M_4, M_5, M_6, M_7 and M_8, respectively, with a suffix [D1]. The results of Table 5 show that EfficientNetV2-S achieves the highest validation accuracy, highlighting its superior classification capability on the dual augmented dataset. The models VGG-16 ($M_6[D1]$), VGG-19 ($M_7[D1]$), and AlexNet ($M_8[D1]$) also demonstrate comparable performance as EfficientNetV2-S ($M_1[D1]$), achieving better validation results in our case. In contrast, other models MobileNetV2 ($M_2[D1]$), DenseNet-201 ($M_3[D1]$), ResNet-50 ($M_4[D1]$) and GoogLeNet ($M_5[D1]$) exhibit relatively lower accuracy, leading to a higher misclassification rate.

The loss versus epoch curves for models $M_1[D1]$ and $M_5[D1]$ are illustrated in Fig. 10. These sample plots also illustrate the convergence characteristics of the respective networks. The EfficientNetV2-S converges more rapidly and

Table 3 Number of images in the training and validation datasets

Dataset configuration	Total images	Plus/No plus	Training images	Validation images
Original dataset (No augmentation)	222	106/116	178	44
Geometric augmented dataset	1272	636/636	1018	254
Dual-stage augmented dataset	3812	1906/1906	3050	762

Table 4 Validation accuracy (%) of pretrained CNNs trained using different augmentation strategies

Networks	No augmentation (222 images)	Validation accuracy of database (%) Only geometric augmentation (1272 images)	Dual-stage augmentation (3812 images)
EfficientNetV2-S	88.60	98.80	99.99
MobileNetV2	81.80	95.10	96.60
DenseNet-201	84.10	95.00	97.00
ResNet-50	86.40	96.40	98.05
GoogLeNet	81.80	96.40	97.60
VGG-16	86.40	98.20	99.10
VGG-19	86.40	97.00	99.00
AlexNet	84.10	98.00	99.10

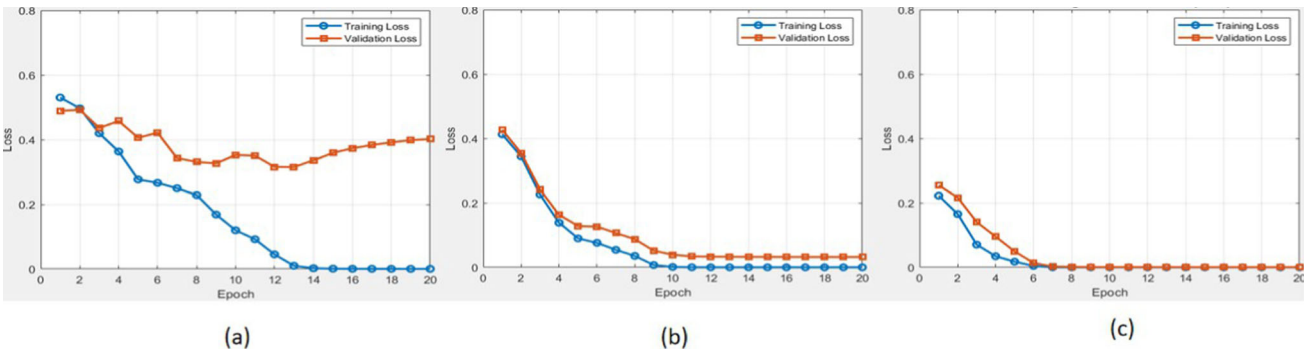


Fig. 9 Loss versus epoch graph of EfficientNetV2-S trained on **a** Original 222 images **b** Geometrical augmented 1272 images **c** Dual stage augmented 3812 images

Table 5 Validation results of pretrained CNNs trained on the complete dataset (D1)

Networks	Base models	Acc (%)	Sens (%)	Validation Spec (%)	Precision (%)	F1-score (%)	AUC (%)
EfficientNetV2-S	$M_1[D1]$	**99.99**	**99.20**	**100**	**100**	**99.60**	**99.61**
MobileNetV2	$M_2[D1]$	96.60	98.50	91.63	92.21	95.71	95.54
DenseNet-201	$M_3[D1]$	97.00	98.70	99.00	99.00	99.28	99.01
ResNet-50	$M_4[D1]$	98.05	98.07	99.00	99.07	99.08	99.00
GoogLeNet	$M_5[D1]$	97.60	97.15	98.02	98.14	97.63	97.64
VGG-16	$M_6[D1]$	**99.10**	**98.00**	**98.00**	**98.23**	**99.07**	**99.04**
VGG-19	$M_7[D1]$	**99.00**	**98.10**	**98.05**	**99.01**	**99.08**	**99.02**
AlexNet	$M_8[D1]$	**99.10**	**99.00**	**99.00**	**99.00**	**99.01**	**99.02**

The bold values are used to highlight the better result among the reported values

Table 6 Test results on a test dataset of 154 images (T1)

Base model	Dataset of 154 images (T1)		
	ACC (%)	Sens (%)	Spec (%)
$M_1[D1]$	**98.1**	**98.7**	**97.4**
$M_2[D1]$	80.3	79.4	63.9
$M_3[D1]$	87.3	87.4	77.9
$M_4[D1]$	88.3	87.4	81.6
$M_5[D1]$	88.0	85.4	82.6
$M_6[D1]$	**90.3**	**97.4**	**82.9**
$M_7[D1]$	**90.3**	**96.2**	**84.2**
$M_8[D1]$	**95.5**	**96.2**	**94.7**

The bold values are used to highlight the better result among the reported values

attains a lower final validation loss compared to the other models. Furthermore, EfficientNetV2-S exhibits a strong overlap between its training and validation loss curves, indicating effective learning and robust generalization. The behavior suggests better optimization and a reduced risk of both overfitting and underfitting.

The performance evaluation of all the models was done on the test data set T1 to validate the models further and check the generalization performance. The results in terms of accuracy, sensitivity, and specificity are summarized in Table 6. The table shows a similar performance that was achieved for the validation results with models $M_1[D1]$, $M_6[D1]$, $M_7[D1]$, and $M_8[D1]$ exhibiting superior generalization performance on the test set compared to models $M_2[D1]$, $M_3[D1]$, $M_4[D1]$, and $M_5[D1]$.

As stated earlier, the primary objective of this work is to develop a Plus disease classifier with zero misclassifications, which can be used by the clinicians to identify all the treatment-requiring cases. Table 6 reveals that although model $M_1[D1]$ outperforms the others across various metrics, it still exhibits some misclassifications. So, to enhance the model's performance, we experimented with various ensembling techniques to check whether the performance of the classifier can be improved.

3 Results

This section describes the analysis done to arrive at the best ensembling model and the results obtained for the given classification problem. Analysis was done using various ensembling approaches using the best-performing base models arrived from the results of part 2.2.

Ensemble classifiers in the literature follow four primary strategies to tap the performance of individual models. Flat stacking [35, 36] (A) employs multiple distinct models on the same dataset, homogeneous ensemble with model variations (B) utilizes variations of a single model architecture trained on the same dataset, a heterogeneous ensemble with data partitioning (C) applies different model architectures to different data partitions and bagging (D) uses training multiple instances of the same model on boot-strapped data subsets. A preliminary analysis with various ensembling approaches gave promising directions with approaches A and D only, and hence the analysis done with flat stacking (A) and bagging (D) is presented here.

Flat Stacking: Flat stacking uses the same data set to train multiple models to take advantage of the model diversity while ensemble learning. Since we have multiple distinct models trained on the complete data set D1, flat stacking was adopted as the ensemble learning strategy. Two ensemble fusion techniques were used for combining the output from the various models via Soft Voting (i.e., averaging of predicted probabilities) and Hard Voting (i.e., majority or max voting) [37]. In the Soft Voting approach, the softmax probability outputs generated by the individual CNN models were averaged at the decision level, and the final predicted class was assigned based on the maximum averaged

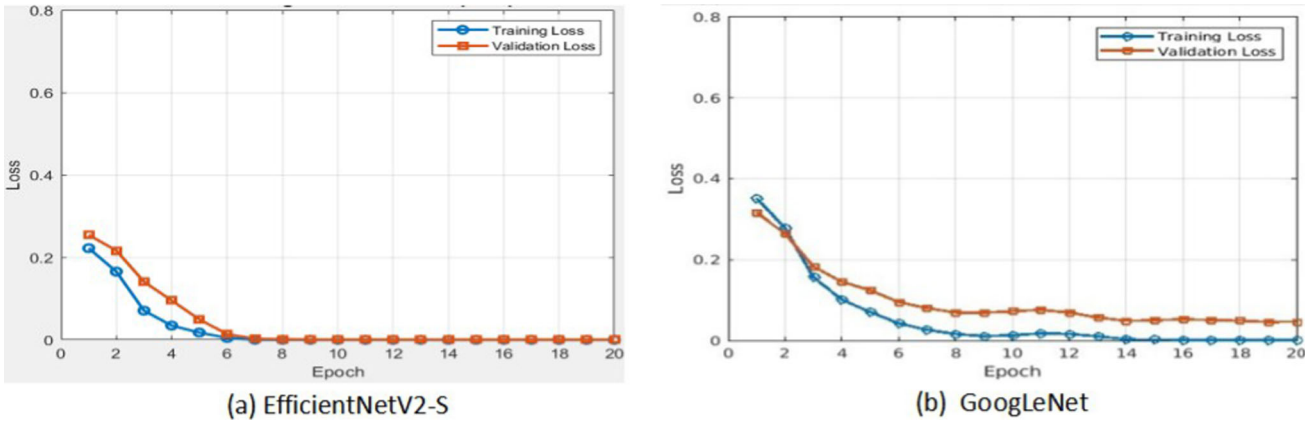


Fig. 10 Loss versus epoch graph of deep convolutional neural networks trained on complete dataset (D1) for **a** EfficientNetV2-S **b** GoogLeNet

Table 7 Ensemble learning using flat stacking on a test dataset of 154 images (T1)

Combinations of base models	No. of base models	Soft voting			Hard voting		
		Acc (%)	Sens (%)	Spec (%)	Acc (%)	Sens (%)	Spec (%)
$M(1,6)[D1]$	2	95.04	98.00	94.07	94.07	94.90	98.00
$M(1,7)[D1]$	2	94.50	97.30	90.80	95.05	93.06	97.40
$M(1,8)[D1]$	**2**	**98.10**	**98.70**	**97.40**	**96.10**	**93.06**	**98.70**
$M(6,7)[D1]$	2	92.20	97.40	86.08	94.08	94.09	94.07
$M(6,8)[D1]$	2	93.50	96.20	90.80	96.10	96.20	96.10
$M(7,8)[D1]$	2	94.02	94.07	90.08	95.50	94.90	96.10
$^{M}(1,6,7)^{[D1]}$	3	95.50	98.70	92.10	95.50	98.70	92.10
$^{M}(6,7,8)^{[D1]}$	3	94.20	96.20	92.10	94.20	96.20	92.10
$^{M}(1,7,8)^{[D1]}$	3	96.10	97.40	94.70	96.10	97.40	94.70
$^{M}(1,6,8)^{[D1]}$	3	96.80	97.40	96.10	96.10	97.40	94.40
$^{M}(1,6,7,8)^{[D1]}$	4	95.50	97.40	93.40	96.10	96.20	96.10

The bold values are used to highlight the better result among the reported values

Table 8 Validation results of pretrained CNNs trained on OD centered datasets (D2)

Networks	Base models	Validation					
		Acc (%)	Sens (%)	Spec (%)	Precision (%)	F1-score (%)	AUC (%)
EfficientNetV2-S	$M_1[D2]$	**99.80**	**99.21**	**99.20**	**99.17**	**99.38**	**99.38**
MobileNetV2	$M_2[D2]$	95.50	98.00	96.00	93.00	96.01	96.01
DenseNet-201	$M_3[D2]$	96.01	99.10	99.00	99.01	99.02	99.00
ResNet-50	$M_4[D2]$	98.00	98.03	98.80	98.76	98.54	98.55
GoogLeNet	$M_5[D2]$	97.00	90.09	99.00	98.10	94.81	95.02
VGG-16	$M_6[D2]$	**98.20**	**98.06**	**98.30**	**98.36**	**98.97**	**98.96**
VGG-19	$M_7[D2]$	**99.00**	**98.02**	**98.06**	**99.02**	**99.08**	**99.08**
AlexNet	$M_8[D2]$	**99.10**	**99.00**	**97.09**	**97.97**	**98.97**	**98.96**

The bold values are used to highlight the better result among the reported values

probability score. In the Hard Voting approach, the final class label was determined based on the majority vote obtained from the constituent CNN models.

The performance outcomes of various combinations of the identified best base models $M_1[D1]$, $M_6[D1]$, $M_7[D1]$ and $M_8[D1]$ were used for analysis. Table 7 summarizes the results obtained on the test data set T1. From the table, it is clear that the combination of base models $M_1[D1]$ and $M_8[D1]$ coined as $M_{(1,8)}[D1]$ gives better performance compared to other models. Although combinations involving two, three, and four base models were explored, none achieved the desired sensitivity of 100% on the test dataset using flat stacking. This limitation prompted the consideration of alternative ensemble strategies to further enhance diagnostic sensitivity.

Bagging: As the ensemble performance achieved through flat stacking was not satisfactory, bagging was subsequently employed by training the same base models on different data samples, in accordance with approaches discussed in the literature [38]. To segregate the data into appropriate subsets for the implementation of this approach, an analysis of the misclassified data was done manually, and it was found that in most of the misclassified samples, the optic disk was slightly dislocated from the center. Motivated by this observation, the complete dataset was segregated into two subsets: (i) an optic disc (OD) centered dataset consisting of 2,412 images, and (ii) a Non-OD centered dataset comprising 140 images. We coin the two data sets as D2 and D3, respectively, for clarity of explanation in the subsequent sections.

To analyze the performance of the two datasets on the CNN models, the datasets were trained on all the pretrained models. The validation performance of all the CNN models trained on datasets D2 and D3 was analyzed, and the results are summarized in Tables 8 and Table 9, respectively. To have better clarity in presenting the models, we call the CNN models trained on OD-centered datasets with a suffix [D2] and CNN models trained on non-OD-centered datasets with a suffix [D3]. The results show that validation performance is better on training with the optic disk (OD) centered dataset.

Table 9 Validation results of pretrained CNNs trained on non OD centered datasets (D3)

Networks	Base models	Validation					
		Acc (%)	Sens (%)	Spec (%)	Precision (%)	F1-score (%)	AUC (%)
EfficientNetV2-S	$M_1[D3]$	93.30	86.61	90.20	90.02	92.17	93.30
MobileNetV2	$M_2[D3]$	61.02	86.01	37.00	56.05	65.42	62.08
DenseNet-201	$M_3[D3]$	74.40	53.40	91.00	88.00	66.00	73.05
ResNet-50	$M_4[D3]$	72.00	53.00	90.00	88.00	66.50	73.95
GoogLeNet	$M_5[D3]$	70.00	40.10	91.20	88.00	57.00	70.40
VGG-16	$M_6[D3]$	61.00	66.00	56.02	58.90	62.35	61.45
VGG-19	$M_7[D3]$	74.00	53.33	93.45	88.88	66.00	73.30
AlexNet	$M_8[D3]$	64.50	80.00	50.20	60.00	98.57	65.00

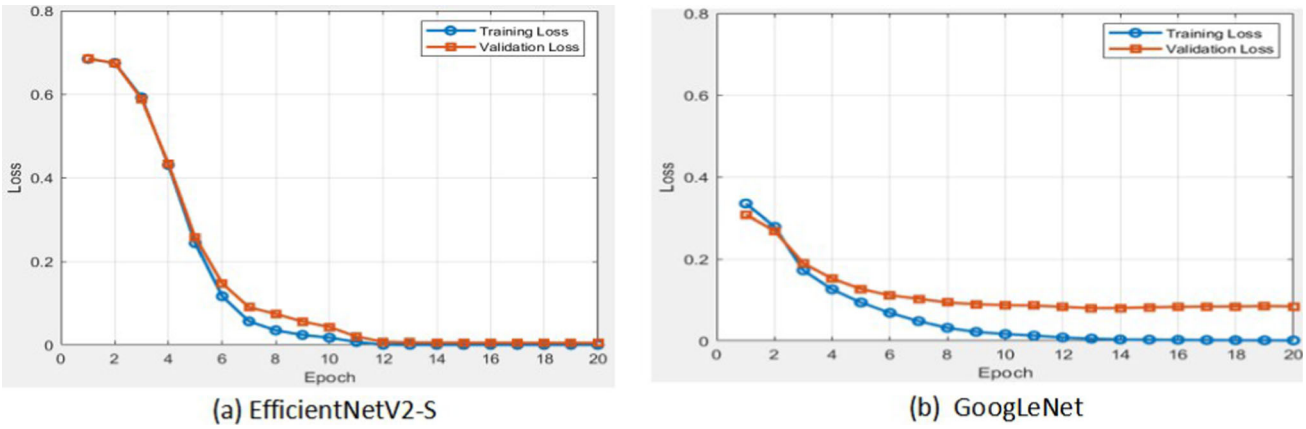


Fig. 11 Loss versus epoch graph of deep convolutional neural networks trained on OD centered Dataset (D2) for **a** EfficientNetV2-S **b** GoogLeNet

The validation performance of networks trained on the non-OD-centered dataset is significantly lower compared to those trained on the complete dataset and OD-centered dataset.

This reduced performance may be attributed to variations in retinal field coverage, reduced visibility of diagnostically relevant vascular structures, illumination inconsistencies, and the limited number of non-OD-centered training samples. In addition, challenging cases such as poor illumination, motion blur, low-contrast retinal images, and severely off-centered retinal fundus images may affect the reliability of the Plus disease classification in real-world screening environments.

Therefore, for subsequent analyses, we focused only on the models trained with the complete dataset (D1) and the OD-centered dataset (D2), as these datasets provide a sufficient number of samples for effective network training. In addition, the retinal vasculature and optic disk region are more clearly visible in OD-centered images, making them more clinically informative for Plus disease classification.

The loss versus epoch graphs for the advanced CNN models trained on D2 are presented in Fig. 11. From the graphs, it is evident that the same trend is followed in the case of the network trained on D2, with EfficientNetV2-S outperforming all the other architectures.

To analyze the performance of the models trained on D1 and D2 on the test dataset, T1 is also segregated into OD-centered images and non-OD-centered images. Out of the 154 images from T1, 54 images were OD centered, and we call this test set T2. Table 10 shows the results of the models trained on D2 when tested on the complete test set T1 and the OD-centered test set T2. The results show that $M_1[D2]$ outperforms the other models on both T1 and T2. It can also be noted that the model $M_1[D2]$ gave zero misclassifications when tested on T2, at the expense of reduced specificity.

The data segregation and performance analysis with various networks trained on different subsets of data have motivated us to implement the ensemble classifier with bagging using the best-performing networks trained on the complete dataset ($M_1[D1]$, $M_6[D1]$, $M_7[D1]$ and $M_8[D1]$) and the best performing networks trained on the OD centered dataset ($M_1[D2]$, $M_6[D2]$, $M_7[D2]$ and $M_8[D2]$).

The bagging approach is applied to the four best-performing networks, which are trained on D1 and D2, with the objective of enhancing classification performance on both T1 and T2. The corresponding performance outcomes for

Table 10 Test results on complete dataset (T1) and OD centered dataset (T2)

Base models	Complete dataset (T1)			OD centered dataset (T2)		
	Acc (%)	Sens (%)	Spec (%)	Acc (%)	Sens (%)	Spec (%)
$M_1[D2]$	**94.8**	**96.2**	**93.4**	**96.3**	**100**	**92.6**
$M_2[D2]$	80.0	86.2	85.7	84.0	90.2	86.7
$M_3[D2]$	85.0	88.2	89.7	88.0	88.2	89.7
$M_4[D2]$	89.0	90.2	89.7	90.0	90.2	89.7
$M_5[D2]$	85.2	92.6	77.8	88.2	92.6	80.8
$M_6[D2]$	**93.0**	**96.2**	**92.7**	**92.6**	**92.6**	**92.6**
$M_7[D2]$	**91.5**	**95.4**	**89.5**	**95.3**	**95.0**	**92.6**
$M_8[D2]$	**92.4**	**95.0**	**92.7**	**94.3**	**96.2**	**92.6**

The bold values are used to highlight the better result among the reported values

Table 11 Ensemble learning using bagging on both datasets

Test dataset	Ensemble network	Soft voting			Hard voting		
		Acc (%)	Sens (%)	Spec (%)	Acc (%)	Sens (%)	Spec (%)
Complete dataset (T1)	$M_1[D1, D2]$	**98.70**	**100.0**	**97.70**	94.80	93.60	96.10
	$M_6[D1, D2]$	90.90	96.20	85.50	91.50	91.20	90.80
	$M_7[D1, D2]$	95.50	97.40	93.40	93.40	92.20	93.70
	$M_8[D1, D2]$	94.80	96.20	93.40	93.10	93.90	92.40
OD centered dataset (T2)	$M_1[D1, D2]$	**100.0**	**100.0**	**100**	98.10	96.30	98.00
	$M_6[D1, D2]$	92.60	91.00	85.20	91.10	91.00	92.30
	$M_7[D1, D2]$	96.30	93.00	92.60	94.40	92.60	96.30
	$M_8[D1, D2]$	92.60	92.60	92.60	94.40	92.60	96.30

The bold values are used to highlight the better result among the reported values

the best model pairs are reported in Table 11. As observed in these tables, the ensemble of EfficientNetV2-S model trained on D1 and D2, $M_1[D1]$ and $M_1[D2]$ ($M_1[D1, D2]$) using the soft voting fusion technique achieved the best results, with an accuracy of 98.7%, sensitivity of 100.0%, and specificity of 97.7% when tested on the complete dataset (T1). This meets our goal of developing a Plus disease classifier with zero misclassifications, a critical requirement for a deployable ROP screening system. We name this proposed model as EfficientNetV2-S BE.

When testing the models on test set T2, the ensemble network attained perfectly zero classification errors with a performance of 100% accuracy, sensitivity, and specificity. These zero-misclassification results set a new benchmark for Plus disease classification on OD-centered ROP images. A visual summary of the performance is provided in the bar plot shown in Fig. 12.

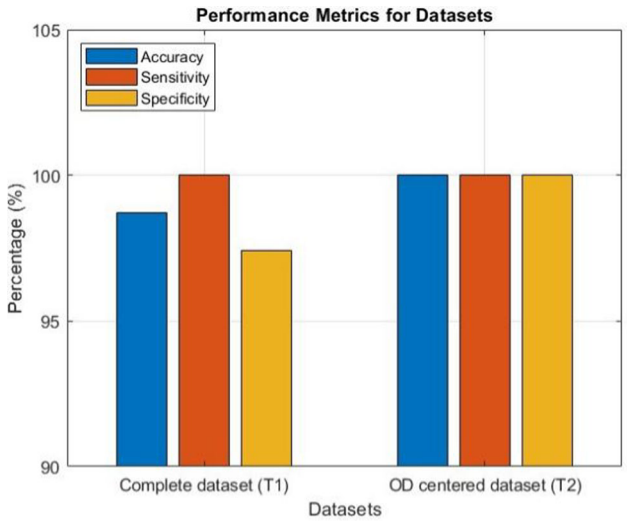


Fig. 12 Performance bar plot of proposed Ensemble Network

4 Discussion

Enhanced performance was observed using the bagging ensemble classifier with zero false positives, using EfficientNetV2-S in combination with the soft voting ensemble technique (average of probabilities), as described in Algorithm 1. This improved performance may be attributed

Table 12 McNemar's Test *p* values for different ensemble models using soft voting ensembling algorithm

Ensemble network	*p* value	
	Complete dataset (T1)	OD centered dataset (T2)
$M_1[D1, D2]$	0.0479	0.0253
$M_6[D1, D2]$	0.1967	0.3171
$M_7[D1, D2]$	0.0736	0.7054
$M_8[D1, D2]$	0.0955	0.4141

to the complementary nature of the individual models, especially in terms of the differences in their error patterns [39]. To statistically evaluate this complementarity among the selected base learners and the effectiveness of the proposed ensemble framework, McNemar's statistical test [40] was employed. This test assesses whether there is a significant difference in the misclassification patterns of two models and whether one model captures complementary information relative to the other, thereby supporting the rationale for ensemble learning. The p value derived from this test indicates the likelihood that the two models behave similarly. $P < 0.05$ implies that the difference in classification outcomes is statistically significant, suggesting that the models complement one another and that ensemble learning is beneficial.

Conversely, $P \geq 0.05$ indicates no significant difference in classification behavior, implying limited benefit from ensemble integration.

As presented in Table 12, the p value for the complete dataset (T1) is 0.0479 and for the OD centered dataset (T2) is 0.0253 in the case of the model pair $M_1[D1, D2]$. Both values are below the 0.05 threshold, leading to the rejection of the null hypothesis at the 5% significance level. This indicates that $M_1[D1]$ and $M_1[D2]$ do not exhibit identical misclassification patterns on either T1 or T2. Instead, their errors occur on different samples, implying a complementary relationship between the two models. Consequently, we conclude that an ensemble of $M_1[D1]$ and $M_1[D2]$ may be beneficial, as they contribute complementary information. Notably, the OD-centered dataset yields a lower p value compared to the complete dataset, suggesting that the divergence in model behavior is more pronounced for T2. Although complementarity is still present in T1, it is to a lesser extent. In contrast, for the remaining model combinations, the null hypothesis could not be rejected, indicating no significant difference in misclassification behavior. Therefore, ensemble learning in those cases does not confer any advantage, which aligns with the performance trends observed in Table 11.

Algorithm 1 Ensemble algorithm

Input: Set of pretrained models M_1, M_2, test dataset D_{test}= D_1, D_2.

Output: Final predicted class labels for D_{test}.

1: Preprocess the input dataset D_{test} to match the required input size of the models.

2: Normalize the data as needed to match the training conditions of the models.

3: For each model M_i,i=(1,2):

- Load the pretrained model M_i
- Feed D_{test} to M_i to obtain the probability predictions P_i for each class.
- P_i=M_i(D_{test}), where P_i is a matrix of probabilities for each class.

4: Compute the average probability matrix P_{avg} across all models:

$P_{avg}(x, c) = \frac{1}{n}\sum_{i=1}^{n} p_i(x, c)$, where x is a sample in D_{test}, c is a class, and n is the number of models.

5: For each sample $x \in D_{test}$:

- Assign the final predicted class label as the one with the highest average probability:

$Y_{pred}(x) = argmax_c P_{avg}(x, c)$

6: Return the final predicted class labels Y_{pred}

The performance of the proposed method is compared with the existing methods for ROP classification. A direct comparison is not possible since there are wide variations in terms of the number of images, the type of architectures employed, and the labeling of classes. However, Table 13 provides a broad comparison based on the classification accuracy, sensitivity, specificity, precision, F-measure, and AUC as metrics. The comparisons are intended primarily to provide a qualitative overview of the existing literature and to position the proposed framework relative to previously

Table 13 Comparison of performance of the proposed method with existing techniques

Study	Images	Label	Classifier	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision	F-score	AUC
Jomier et al. [41]	20	Plus/Pre-Plus/Normal	Neural network	–	80	82	–	–	–
Wallace et al. [42]	20	Plus/No Plus	Threshold-based decision rule	–	80	91	–	–	–
Wittenberg [43]	77	Plus/No Plus	SVM	85	87	82	–	–	–
Ataer [44]	77	Plus/Pre-Plus	SVM	95	–	–	–	–	–
Oloumi [45]	110	Plus/No Plus	Random tree	76	73	66	–	–	0.76
Rani [46]	56	ROP/Normal	KNN	89.64	–	–	–	–	–
Pour [47]	87	Plus/Pre-Plus/Normal	KNN	72.3	–	–	–	–	–
Zamani [47]	87	Plus/Pre-Plus/Normal	SVM	83.7	–	–	–	–	–
Mahmoudi [47]	87	Plus/No Plus/Preplus	MLP	84.4	–	–	–	–	–
Kadge et al. [5]	60	ROP/Normal	KNN	85	–	—	–	–	–
Samant [5]	60	ROP/Normal	SVM	92.5	–	–	–	–	–
Nisha [23]	178	Plus/No Plus	SVM	93	95	93	–	–	–
Coyner [48]	579	ROP/Normal	Logistic regression	–	–	80.8	–	–	–
Sankari [49]	200	ROP/Normal	Quantum SVM	95.5	93	–	–	–	0.97
Yenice [50]	280	ROP/Normal	X- Tree	–	–	84	–	–	–
Jemshi [51]	178	Plus/No Plus	Neural Network	96.3	100	93.8	–	–	–
Worrall et al. [8]	1459	ROP/Normal	GoogLeNet	93.6	95.4	94.6	0.713	–	–
Wang et al. [9]	2361	ROP/Normal	Two-stage DNN(Id-Net + Gr-Net)	95	96.64	–	–	–	–
Brown et al. [52]	5511	Plus/Pre-Plus/Normal	GoogLeNet	–	93	94	–	–	0.94
Hu Jnjie et al. [53]	2668	ROP/Normal	Inception-V2	–	96	–	–	–	–

Table 13 (continued)

Study	Images	Label	Classifier	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision	F-score	AUC
Mao et al. [10]	3311	Plus/Pre-Plus/Normal	DenseNet	93.31	95.1	–	–	–	0.9325
Zhang et al. [13]	17801	ROP/Normal	VGG-16		94	—	93	–	–
Tong et al. [11]	36,231	Normal/Urgent	ResNet-101,Faster- RCNN	90.3	77.8	93.2	–	0.761	–
Lei et al. [12]	17801	ROP/Normal	Modified ResNet-50	–	98.03	94.55	–	–	–
Coyner et al. [55]	5842	Plus/Pre-Plus/Normal	ResNet-18	–	–	–	–	–	0.97
WuQ et al. [54]	7033	ROP/Normal	ResNet-50	–	–	–	–	–	0.94
Luo et al. [58]	9000	ROP/Normal	ResNet-101	75	–	–	–	–	0.60
Zhang et al. [56]	10521	ROP/Normal	Deep attentive CNN		93.09	87.70	–	–	–
Rao et al. [57]	227326	ROP/Normal	EfficientNet- B0	–	91	91.22	–	–	–
Nazar et al. [14]	2776	ROP/Normal	ResNet50, DenseNet161, EfficientNetB5	90.28	–	–	–	–	–
Elizabeth et al. [15]	11191	ROP/Normal/Stage 2/3	Customized ResNet-50,DNN	92.8	94.9	–	–	–	–
Raja et al. [16]	2400	ROP/Normal	Quantum mobile vision transformer	96.67	–	–	–	–	–
Agrawal et al. [17]	191	Plus/Pre-Plus/Normal	Inceptionv3,Xception,ResNet152	87	–	–	–	–	–
Nasser et al. [18]	1700	Normal/Urgent	MobileNet	91.39	94.9	–	–	–	–
Esraa et al. [19]	7637	Plus/Pre-Plus/Normal	ResNet50,EfficientNet-B4	96.9	–	–	–	–	–
Proposed (On T1)	**3966**	**Plus/No Plus**	**EfficientNetV2-S BE**	**98**	**100**	**97**	**1**	**98.1**	**0.98701**
Proposed (On T2)	**3866**	**Plus/No Plus**	**EfficientNetV2-S BE**	**100**	**100**	**100**	**100**	**100**	**1**

The bold values are used to highlight the better result among the reported values

reported approaches, rather than to claim direct one-to-one performance superiority.

The comparison indicates that the proposed method achieves competitive or superior performance compared to previously reported approaches in terms of accuracy, sensitivity, precision, F-score, and AUC. The third column in the table defines the labels of classification, and it can be seen that most of the deep learning models in the literature, go for an ROP/Normal classification. An accurate binary classification of the Plus/No Plus cases, which is a major benchmark for identifying the treatment-requiring ROP cases in an ROP screening system, is not addressed anywhere in the literature. This makes our method novel and promising for an ideal ROP screening system.

The proposed framework is intended as a computer-aided screening support tool for retinopathy of prematurity (ROP) rather than a fully autonomous diagnostic system. Furthermore, the proposed model achieves inference in the order of milliseconds per image on the specified GPU configuration with modest computational requirements, demonstrating its suitability for real-time deployment. The high sensitivity achieved by the proposed framework is clinically important in screening applications, where minimizing false negative predictions is essential for timely referral and treatment. The system may assist ophthalmologists by providing rapid preliminary screening and reducing workload in resource-limited clinical settings.

In a practical clinical workflow, retinal images can be captured by trained technicians at peripheral centers and analyzed by the proposed system, enabling ophthalmologists to remotely review and prioritize cases. Consequently, the frame-work is well-suited for integration into real-time clinical screening applications. Nevertheless, further validation using larger publicly available and multi-center retinal fundus datasets would further strengthen the robustness, generalizability, and clinical applicability of the proposed framework, particularly for non-OD-centered retinal images acquired under diverse clinical conditions.

5 Conclusion

One of the prolonged requirements from the ROP experts is a widely deployable, sustainable, and cost-effective retinopathy of prematurity (ROP) screening system capable of accurately identifying both discharge cases and those requiring treat-ment. A computer-aided diagnostic (CAD) system achieving 100% sensitivity would be ideally suited for this purpose. The proposed work addresses this requirement by developing an ROP screening system with zero false negatives, ensuring that no diseased cases are left untreated. The proposed work also addresses another major challenge of data scarcity in medical image analysis. To mitigate this, the proposed work uses a transfer learning approach to address the problem on one side and also proposes an augmentation strategy that employs enhanced images to train the deep networks on the other side. The proposal is more general, where the enhancement methods employed can be specific to a given problem, depending on the details that are to be learned by the network. Through extensive experimentation and comprehensive evaluation, EfficientNetV2-S has been identified as the most effective state-of-the-art Convolutional Neural Network (CNN) architecture for the classification of Plus disease in ROP. The analysis done on the available data set to split the data effectively into two complementary datasets is another contribution of the work. Finally, the goal of 100% sensitivity could be achieved with a novel ensemble framework composed of the best-performing CNNs, selected through a rigorous analysis. The achievement of 100% sensitivity and specificity for the optic disc (OD) centered images is highly promising, provided a lightweight algorithm could be developed to automatically segregate OD-centered images from captured datasets.

The automated image analysis systems developed through this research can greatly assist ophthalmologists in the timely diagnosis and treatment of ROP. These systems are especially valuable in remote and economically disadvantaged regions. In such environments, the technicians can capture and upload the images in real time, and the decision from the proposed screening system, along with the data, can be sent to the hospital for further verification by the clinicians. The proposed algorithm can be easily integrated with existing telemedicine programs like KIDROP, eliminating the need for an in-person examination by specialized ROP experts.

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Data Availability The datasets generated during and/or analyzed during the current study are not publicly available due to ethical/legal/commercial restrictions.

Declarations

Conflict of interest The authors declare they have no financial interests.

Ethical approval This study is approved by the institutional review board and ethics committee of Narayana Netralaya Eye Hospital, Bangalore, India.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publications The authors affirm that parents of human research participants provided informed consent for publication of the images.

References

1. Le, C.; Basani, L.B.; Zurakowski, D.; Ayyala, R.S.; Agraharam, S.G.: Retinopathy of prema-turity: incidence, prevalence, risk factors, and outcomes at a tertiary care center in Telangana. J. Clin. Ophthalmol. Res. **4**(3), 119–122 (2016)
2. Sommer, A.; Taylor, H.R.; Ravilla, T.D.; West, S.; Lietman, T.M.; Keenan, J.D.; Chiang, M.F.; Robin, A.L.; Mills, R.P.: Challenges of ophthalmic care in the develop-ing world. JAMA Ophthalmol. **132**(5), 640–644 (2014)
3. Nisha, K.L.; Ganapathy, S.; Puthumangalathu Savithri, S.; Idaguri, M.; Mohanachandran, P.; Vinekar, A.; Chandra, P.; Kulkarni, S.; Dogra, M.: A novel method to improve inter-clinician variation in the diagnosis of retinopathy of pre-maturity using machine learning. Curr. Eye Res. **48**(1), 60–69 (2023)
4. Vinekar, A.: IT-enabled innovation to prevent infant blindness in rural India: the KIDROP experience. J. Indian Bus. Res. **3**(2), 98–102 (2011)
5. Samant, A.; Kadge, S.: Classification of a reti-nal disease based on different supervised learn-ing techniques. Int. J. Sci. Res. Netw. Secur. Commun. **5**(3), 9–13 (2017)
6. Chiang, M.F.; Gelman, R.; Jiang, L.; Martinez-Perez, M.E.; Du, Y.E.; Flynn, J.T.: Plus disease in retinopathy of prematurity: an analysis of diagnostic performance. Trans. Am. Ophthalmol. Soc. **105**, 73 (2007)
7. Tafsast, A., Hadjili, M.L., Hafdaoui, H., Bouakaz, A., Benoudjit, N.: Automatic Gaus-sian mixture model (GMM) for segmenting 18F-FDG-PET images based on Akaike information criteria. In 2015 4th Int. Conf. Electr. Eng. 1–4 (2015)
8. Worrall, D.E., Wilson, C.M., Brostow, G.J.: Automated retinopathy of prematurity case detection with convolutional neural networks. In: International Workshop on Deep Learning in Medical Image Analysis. Springer International Publishing, Cham. 68–76 (2016)
9. Wang, J.; Ju, R.; Chen, Y.; Zhang, L.; Hu, J.; Wu, Y.; Dong, W.; Zhong, J.; Yi, Z.: Automated retinopathy of prematurity screening using deep neural networks. EBioMedicine **35**, 361–368 (2018)
10. Mao, J.; Luo, Y.; Liu, L.; Lao, J.; Shao, Y.; Zhang, M.; Zhang, C.; Sun, M.; Shen, L.: Auto-mated diagnosis and quantitative analysis of plus disease in retinopathy of prematurity based on deep convolutional neural networks. Acta Ophthalmol. **98**(3), e339–e345 (2020)
11. Tong, Y.; Lu, W.; Deng, Q.Q.; Chen, C.; Shen, Y.: Automated identification of retinopathy of prematurity by image-based deep learning. Eye Vis. **7**, 40 (2020)
12. Lei, B.; Zeng, X.; Huang, S.; Zhang, R.; Chen, G.; Zhao, J.; Wang, T.; Wang, J.; Zhang, G.: Automated detection of retinopathy of prema-turity by deep attention network. Multimed. Tools Appl. **80**, 36341–36360 (2021)
13. Zhang, R.; Zhao, J.; Xie, H.; Wang, T.; Chen, G.; Zhang, G.: Automatic diagnosis for aggressive posterior retinopathy of prematurity via deep attentive convolutional neural network. Expert Syst. Appl. **187**, 115843 (2022)
14. Salih, N.; Ksantini, M.; Hussein, N.; Halima, D.B.; Abdul Razzaq, A.; Ahmed, S.: Deep learn-ing models and fusion classification technique for accurate diagnosis of retinopathy of pre-maturity in preterm newborn. Baghdad Sci. J. **21**(5), 21 (2024)
15. Ndunge Mutua, E.; Shibwabo Kasamani, B.; Reich, C.: Retinopathy of prematurity disease diagnosis using deep learning. Int. J. Comput. Digit. Syst. **16**(1), 1097–1110 (2024)
16. Umapathy, S.; Chandrasekaran, A.; Baskaran, P.; Dhanraj, V.: Automated detection of stages, zones, and plus diseases of retinopathy of prematurity using quantum convolutional networks in neonatal fundus images. Eng. Appl. Artif. Intell. **160**, 111938 (2025)
17. Agrawal, R.; Kulkarni, S.; Deshpande, M.; Gaikwad, A.; Walambe, R.; Kotecha, K.V.: Classification of Stages 1, 2, 3 and Preplus, Plus disease of ROP using MultiCNN LSTM classifier. MethodsX. **14**, 103182 (2025)
18. Shoeibi, N.; Ameri, N.; Hoseinkhani, M.R.; Ansari Astaneh, M.R.; Motamed Shariati, M.; Hosseini, S.M.; Abrishami, M.; Abrishami, M.; Zamani, G.; Heidarzadeh, H.R.: Deep learning algorithms for timely diagnosis of retinopathy of prematurity requiring treatment. Eye **40**(1), 63–70 (2026)
19. Mohiy, E.; AbdulWakel, H.I.; Khairy, M.; Houssein, E.H.: An efficient deep learning model for reliable detection and classification of retinopathy of prematurity. Discov. Artif. Intell. **6**, 305 (2026)
20. Mohammed, A.; Kora, R.: A comprehensive review on ensemble deep learning: opportunities and challenges. J. King. Saud. Univ. Comput. Inf. Sci. **35**(2), 757–774 (2023)
21. Chen, Y.; Wang, Y.; Gu, Y.; He, X.; Ghamisi, P.; Jia, X.: Deep learning ensemble for hyper-spectral image classification. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. **12**(6), 1882–1897 (2019)
22. Reddy, A.V.N.; Mallick, P.K.; Kumar, S.; Mishra, D.; Reddy, P.A.; Chindam, S.: Meerkat-optimized SENet approach: advancements in retinal fundus image augmentation and clas-sification. Arab. J. Sci. Eng. **50**(19), 15235–15279 (2025)
23. Nisha, K.L.; Sreelekha, G.; Sathidevi, P.S.; Mohanachandran, P.; Vinekar, A.: A computer-aided diagnosis system for plus disease in retinopathy of prematurity with structure adap-tive segmentation and vessel based features. Comput. Med. Imaging Graph. **74**, 72–94 (2019)
24. Yildiz, V.M.; Tian, P.; Yildiz, I.; Brown, J.M.; Kalpathy-Cramer, J.; Dy, J.; Ioanni-dis, S.; Erdogmus, D.; Ostmo, S.; Kim, S.J.; Chan, R.P.; Campbell, J.P.; Chiang, M.F.: Plus disease in retinopathy of prematurity: convolutional neural network performance using a combined neural network and feature extraction approach. Transl. Vis. Sci. Technol. **9**(2), 10 (2020)
25. Huang, Y.P.; Vadloori, S.; Chu, H.C.; Kang, E.Y.C.; Wu, W.C.; Kusaka, S.; Fukushima, Y.: Deep learning models for automated diagnosis of retinopathy of prematurity in preterm infants. Electronics **9**(9), 1444 (2020)
26. Hassan, E.; Ghadiri, H.: Advancing brain tumor classification: a robust framework using EfficientNetV2 transfer learning and statistical analysis. Comput. Biol. Med. **185**, 109542 (2025)
27. Akay, M.; Du, Y.; Sershen, C.L.; Wu, M.; Chen, T.Y.; Assassi, S.; Mohan, C.; Akay, Y.M.: Deep learning classification of systemic sclerosis skin using the MobileNetV2 model. IEEE Open J. Eng. Med. Biol. **2**, 104–110 (2021)
28. Hasan, N.; Bao, Y.; Shawon, A.; Huang, Y.: DenseNet convolutional neural networks application for predicting COVID-19 using CT image. SN Comput. Sci. **2**(5), 389 (2021)
29. Behar, N.; Shrivastava, M.: ResNet50-based effective model for breast cancer classification using histopathology images. Comput. Model. Eng. Sci. **130**(2), 823–839 (2022)
30. Al-Huseiny, M.S.; Sajit, A.S.: Transfer learn-ing with GoogLeNet for detection of lung can-cer. Indones. J. Electr. Eng. Comput. Sci. **22**(2), 1078–1086 (2021)
31. Pardede, J.; Sitohang, B.; Akbar, S.; Khodra, M.L.: Implementation of transfer learning using VGG16 on fruit ripeness detection. Int. J. Intell. Syst. Appl. **13**(2), 52–61 (2021)
32. Mateen, M.; Wen, J.; Song, S.; Huang, Z.: Fundus image classification using VGG-19 architecture with PCA and SVD. Sym-metry. **11**(1), 1 (2019)

33. Singh, I.; Goyal, G.; Chandel, A.: AlexNet architecture based convolutional neural network for toxic comments classification. J. King Saud Univ. - Comput. Inf. Sci. **34**(9), 7547–7558 (2022)
34. Bharathiraja, S.; Rajesh Kanna, B.; Hariharan, M.: A deep learning framework for image authentication: an automatic source camera identification Deep-Net. Arab. J. Sci. Eng. **48**(2), 1207–1219 (2023)
35. Rane, N.; Choudhary, S.P.; Rane, J.: Ensemble deep learning and machine learning: applications, opportunities, challenges, and future directions. Stud. Med. Health Sci. **1**(2), 18–41 (2024)
36. Mohapatra, S.; Maneesha, S.; Mohanty, S.; Patra, P.K.; Bhoi, S.K.; Sahoo, K.S.; Gandomi, A.H.: A stacking classifiers model for detecting heart irregularities and predicting cardiovascu-lar disease. Healthc. Anal. **3**, 100133 (2023)
37. Keerthana, I.P.; Satheesh Kumar, R.: An enhanced transfer learning-based hierarchical ensemble framework for diabetic retinopathy identification and multistage classification. Arab. J. Sci. Eng. **51**, 6865–6895 (2025)
38. Antonio, B.; Moroni, D.; Martinelli, M.: Efficient adaptive ensembling for image classification. Expert. Syst. **42**(1), e13424 (2025)
39. Kundu, R.; Singh, P.K.; Ferrara, M.; Ahma-dian, A.; Sarkar, R.: ET-NET: an ensem-ble of transfer learning models for prediction of COVID-19 infection through chest CT-scan images. Multimed. Tools Appl. **81**(1), 31–50 (2022)
40. Silva, L.F.J.; Cortes, O.A.C.; Diniz, J.O.B.: A novel ensemble CNN model for COVID-19 classification in computerized tomography scans. Results Control Optim. **11**, 100215 (2023)
41. Jomier, J., Wallace, D.K., Aylward, S.R.: Quantification of retinopathy of prematurity via vessel segmentation. In: Medical Image Com-puting and Computer-Assisted intervention-MICCAI 2003: 6th International Conference, montreal, Canada Proceedings Springer Berlin Heidelberg. 620–626 (2003)
42. Wallace, D.K.; Jomier, J.; Aylward, S.R.; Landers, M.B., III.: Computer-automated quan-tification of plus disease in retinopathy of pre-maturity. J. AAPOS **7**(2), 126–130 (2003)
43. Wittenberg, L.A.; Jonsson, N.J.; Chan, R.V.P.; Chiang, M.F.: Computer-based image analysis for plus disease diagnosis in retinopathy of prematurity. J. Pediatr. Ophthalmol. Strabismus **49**(1), 11–19 (2012)
44. Ataer-Cansizoglu, E., Bolon-Canedo, V., Campbell, J.P., Bozkurt, A., Erdogmus, D., Kalpathy-Cramer, J., Patel, S., Jonas, K., Chan, R.V.P., Ostmo, S., Chiang, M.F., the i-ROP Research Consortium.: Computer-based image analysis for plus disease diagnosis in retinopathy of prematurity: performance of the i-ROP system and image features associated with expert diagnosis. Transl. Vis. Sci. Technol. 4(6), 5 (2015)
45. Oloumi, F.; Rangayyan, R.M.; Casti, P.; Ells, A.L.: Computer-aided diagnosis of plus disease via measurement of vessel thickness in retinal fundus images of preterm infants. Comput. Biol. Med. **66**, 316–329 (2015)
46. Rani, P.; Elagiri Ramalingam, R.; Raja-mani, K.T.; Kandemir, M.; Singh, D.: Multiple instance learning: robust validation on retinopathy of prematurity. Int. J. Control Theory Appl. **9**(36), 451–459 (2016)
47. Pour, E.K.; Pourreza, H.; Zamani, K.A.; Mahmoudi, A.; Sadegh, A.M.; Shadravan, M.; Karkhaneh, R.; Pour, R.R.; Esfahani, M.R.: Retinopathy of prematurity-assist: novel software for detecting plus disease. Korean J. Ophthalmol. **31**(6), 524–532 (2017)
48. Coyner, A.S.; Chen, J.S.; Singh, P.; Sch-elonka, R.L.; Jordan, B.K.; McEvoy, C.T.; Anderson, J.E.; Chan, R.V.P.; Sonmez, K.; Erdogmus, D.; Chiang, M.F.; Kalpathy-Cramer, J.; Campbell, J.P.: Single-examination risk prediction of severe retinopathy of prematurity. Pediatrics **148**(6), e2021051772 (2021)
49. Sankari, V.M.R.; Umapathy, U.; Alasmari, S.; Aslam, S.M.: Automated detection of retinopathy of prematurity using quantum machine learning and deep learning techniques. IEEE Access **11**, 94306–94321 (2023)
50. Yenice, E.K.; Kara, C.; Yenice, M.; Erdas, C.B.: Retinopathy of prematurity in late preterm twins with a birth weight discordance: can it be predicted by artificial intelligence. Beyoglu Eye J. **8**(4), 287–292 (2023)
51. Jemshi, K.M.; Sreelekha, G.; Sathidevi, P.S.; Mohanachandran, P.; Vinekar, A.: Plus disease classification in retinopathy of prematurity using transform based features. Multimedia Tools Appl. **83**(1), 861–891 (2024)
52. Brown, J.M.; Campbell, J.P.; Beers, A.; Chang, K.; Ostmo, S.; Chan, R.V.P.; Dy, J.; Erdogmus, D.; Ioannidis, S.; Kalpathy-Cramer, J.; Chiang, M.F.: Automated diagnosis of plus disease in retinopathy of prematurity using deep convolutional neural networks. JAMA Ophthalmol. **136**(7), 803–810 (2018)
53. Hu, J.; Chen, Y.; Zhong, J.; Ju, R.; Yi, Z.: Automated analysis for retinopathy of prema-turity by deep neural networks. IEEE Trans. Med. Imaging **38**(1), 269–279 (2018)
54. Wu, Q.; Hu, Y.; Mo, Z.; Wu, R.; Zhang, X.; Yang, Y.; Liu, B.; Xiao, Y.; Zeng, X.; Lin, Z.; Fang, Y.; Wang, Y.; Lu, X.; Song, Y.; Ng, W.W.Y.; Feng, S.; Yu, H.: Development and validation of a deep learning model to predict the occurrence and severity of retinopathy of prematurity. JAMA Netw. Open **5**(6), e2217447 (2022)
55. Coyner, A.S.; Chen, J.S.; Chang, K.; Singh, P.; Ostmo, S.; Chan, R.V.P.; Chiang, M.F.; Kalpathy-Cramer, J.; Campbell, J.P.: Synthetic medical images for robust, privacy-preserving training of artificial intelligence: application to retinopathy of prematurity diagnosis. Ophthal-mol. Sci. **2**(2), 100126 (2022)
56. Zhang, R.; Zhao, J.; Xie, H.; Wang, T.; Chen, G.; Zhang, G.; Lei, B.: Automatic diagnosis for aggressive posterior retinopathy of prematurity via deep attentive convolutional neural network. Expert Syst. Appl. **187**, 115843 (2022)
57. Rao, D.P.; Savoy, F.M.; Tan, J.Z.E.; Fung, B.P.E.; Bopitiya, C.M.; Sivaraman, A.; Vinekar, A.: Development and validation of an artificial intelligence based screening tool for detection of retinopathy of prematurity in a South Indian population. Front. Pediatr. **11**, 1197237 (2023)
58. Luo, Z.; Ding, X.; Hou, N.; Wan, J.: A deep-learning-based collaborative edge–cloud telemedicine system for retinopathy of prema-turity. Sensors **23**(1), 276 (2023)

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